Learning Spatial Unit-Norm Latents with Riemannian Flow Matching

SRUL: Spherical Riemannian Unit-Norm Latents

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Motivation

An autoencoder does not have to produce Gaussian latents

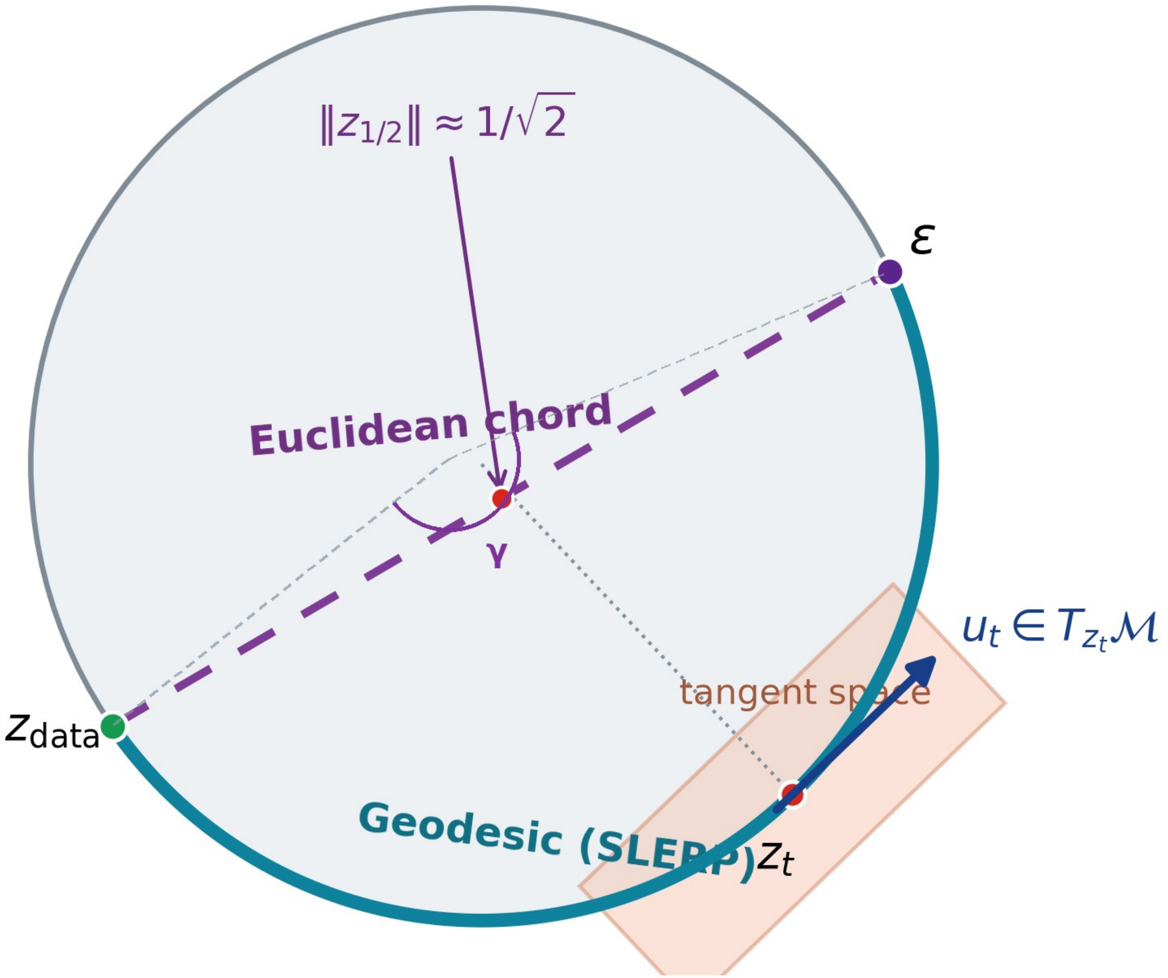
Normalization can make token norms nearly constant
(e.g., Transformer + LayerNorm)

The prior can mismatch the latent support or leave it along the transport path

Main contribution

SRUL learns spatial unit-norm tokens, controls information with tangent noise, and models the same spherical support using RFM.

CIFAR-10 • PathMNIST • CelebA-64



Selected refs: VAE (Kingma & Welling, 2014); LDM (Rombach et al., 2022); Flow Matching (Lipman et al., 2023).

SRUL pipeline and network architecture

CIFAR-10 implementation used for the main geometry and ablation experiments

1 Spatial spherical autoencoder

Image 3×32×32 → **CNN encoder** 3→96→192→384→32, 32×32 → 4×4 → **Unit-norm tokens** 32×4×4, L2 per token → **CNN decoder** 32→384→192→96→3, 4×4 → 32×32

2 Class-conditioned convolutional RFM prior

Sphere noise 32×4×4 → **SLERP state z_t** → **Conv RFM prior** 32→256→32, 6 ResBlocks, 128-D time + class → **Tangent velocity** → **Target u_t**

3 Riemannian ODE sampling and decoding

Noise → **100 Exp-map steps projected velocity** → **Generated latent** 32×4×4 → **Same CNN decoder** → **Image**

Key equations

Unit-norm token

$$z_{i,j} = \frac{h_{i,j}}{\|h_{i,j}\|_2} \in \mathbb{S}^{31}$$

Geodesic training path

$$z_t = \frac{\sin((1-t)\gamma)}{\sin\gamma} z_0 + \frac{\sin(t\gamma)}{\sin\gamma} z_1$$

$$\gamma = \arccos(z_0^\top z_1)$$

Velocity matching

$$\mathcal{L}_{\text{RFM}} = \mathbb{E}\left[\|\Pi_{z_t} v_\theta(z_t, t, y) - u_t\|_2^2\right]$$

Manifold-preserving sampling

$$z_{t+\Delta t} = \mathrm{Exp}_{z_t}(\Delta t\, \Pi_{z_t} v_\theta(z_t, t, y))$$

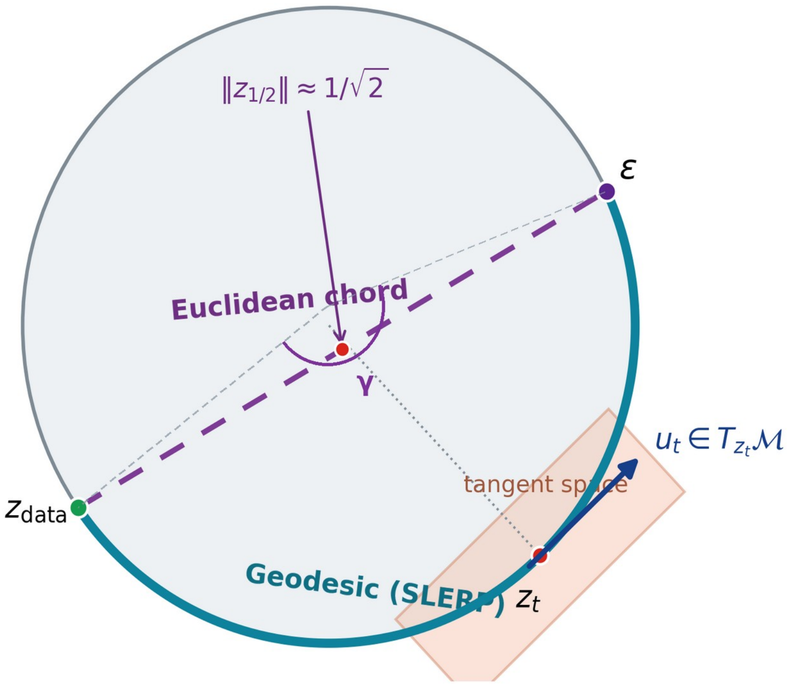
The encoder support, training path, and sampling update use the same geometry.

Method refs: Sphere Encoder (Yue et al., 2026); Flow Matching (Lipman et al., 2023); Riemannian/RJF flow motivation (Kumar & Patel, 2026).

Why the autoencoder and prior must match

Two controlled tests separate path mismatch from support mismatch

Same spherical endpoints, different transport path



1. Path match: same spherical autoencoder

Method	FID	KID	Prec.	Recall	Min norm
Chord	67.44	0.0740	0.840	0.085	0.705
SRUL	**63.84**	**0.0667**	**0.857**	**0.091**	**1.000**

2. Support match: SRUL vs. the same VAE with two priors (CFG = 2)

Method	FID	Prec.	Recall
SRUL: spherical AE + RFM	**49.42**	**0.857**	**0.110**
VAE + standard FM	50.70	0.855	0.091
VAE + projected RFM	56.04	0.848	0.063

Same VAE: radius CV = 0.291; mean-radius replacement changes reconstruction FID 16.90 → 22.46.

SRUL learns spherical support before RFM. Projecting a VAE after training removes radius information.

Baselines and metrics build on VAE/LDM latent modeling (Kingma & Welling, 2014; Rombach et al., 2022) and Flow Matching (Lipman et al., 2023).

Information control and conditional sampling

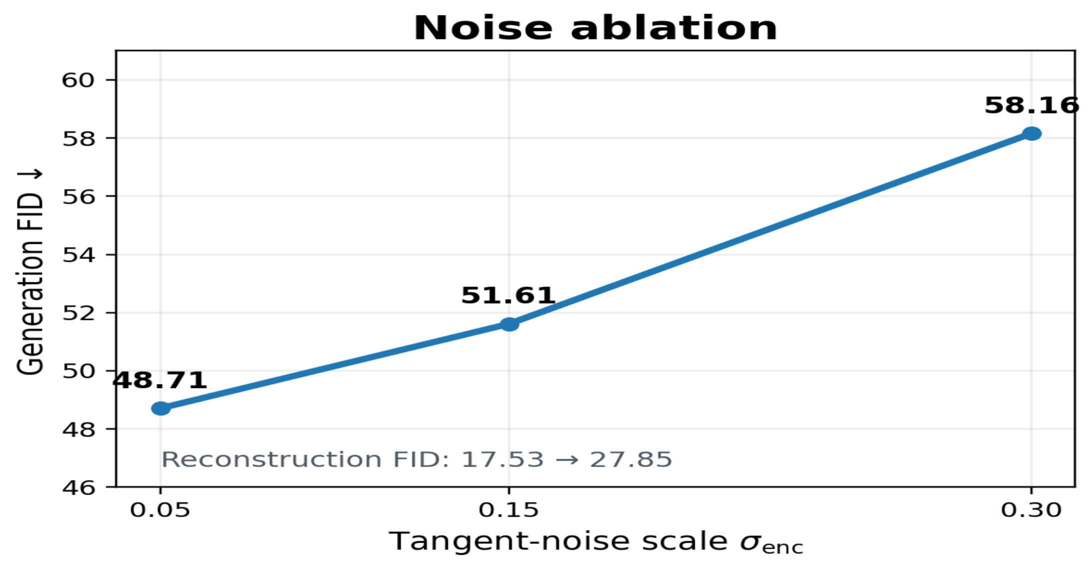
Two separate controlled studies on CIFAR-10: left fixes CFG s = 2; right fixes $\sigma_{en}c = 0.15$

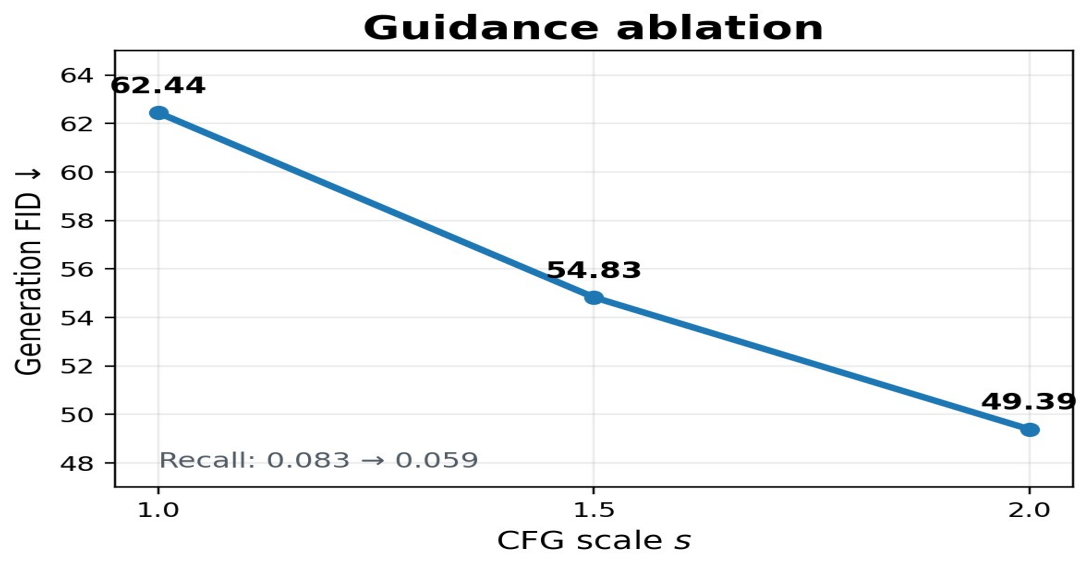
Tangent-noise information control

$$z_\sigma = \mathrm{Exp}_z(\sigma_{\mathrm{enc}}\,\Pi_z(\xi))$$

Classifier-free guidance

$$v_{\mathrm{CFG}} = v_\varnothing + s(v_y - v_\varnothing)$$





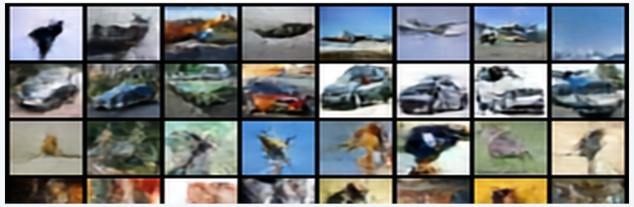
At fixed s = 2, $\sigma_{en}c = 0.30$ degrades all datasets; PathMNIST is slightly best at 0.15.

At fixed $\sigma_{en}c = 0.15$, CIFAR-10 benefits from stronger guidance; PathMNIST peaks at s = 1.5 and CelebA changes little.

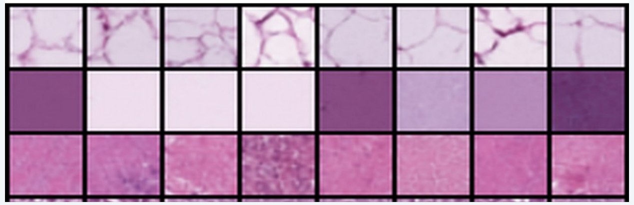
Ablation refs: Unified Latents information-control view (Heek et al., 2026); Classifier-Free Guidance (Ho & Salimans, 2022).

SRUL in three image settings

Same reference setting in every row: σ_{enc} = 0.15 and CFG s = 2



CIFAR-10



PathMNIST

CelebA-64

Dataset	Evaluation setting	Grid	Recon. FID	Gen. FID	Prec.	Recall
CIFAR-10	Geometry + sampling ablations	4 × 4	17.31	49.39	0.884	0.059
PathMNIST	Histopathology transfer	4 × 4	7.13	29.36	0.829	0.309
CelebA-64	64 × 64 face generation	8 × 8	9.08	41.22	0.657	0.198

The same pipeline transfers to natural objects, histopathology, and 64 x 64 faces; the controlled sweeps are reported separately.

Dataset refs: CIFAR-10 (Krizhevsky, 2009); MedMNIST v2 / PathMNIST (Yang et al., 2023); CelebA (Liu et al., 2015).